TRACK: UX/UI DESIGN · WEEK 6 · BUILD

UX-06C: Design System AI Sprint (Prompt to Production)

2-Day Sprint Architecture · Prompt Engineering Log · MoSCoW Matrix · DESIGN.md Summary

Candidate: K. M. Siam
System: FlyCore v1.0

SPRINT ROLE	PROMPT ITERATIONS	TRIAGE FRAMEWORK	CORE DELIVERABLE
AI Design System Architect	4 Documented Cycles	MoSCoW Feedback Matrix	Figma + DESIGN.md + PDF

1. Prompt Engineering Log (Audit of 4 Iteration Cycles)

Iteration 1: Color Tokens & WCAG Contrast

Initial Failure ("AI Slop"): Prompting *"give me colors"* produced disconnected hexes (`#9CA3AF` text on `#F3F4F6` bg) with a 2.4:1 contrast failure (violating WCAG AA) and zero Dark Mode consideration.

Production Refinement: Calibrated prompt strictly enforcing semantic functional names (`surface-*`, `text-*`, `action-*`), dual-mode variables, and mandatory >4.5:1 mathematical contrast validation.

Iteration 2: Typography Scale & Modular Rhythm

Initial Failure ("AI Slop"): DeepSeek generated arbitrary linear 4px increments (32px, 28px, 24px, 20px) resulting in muddy visual hierarchy with zero typographic rhythm.

Production Refinement: Enforced an accessible 16px base scaling on a 1.250 Major Third modular ratio, tabular numerals (`tnum`) for table counts, and unitless line heights.

Iteration 3: Spacing Scale & Warehouse Touch Targets

Initial Failure ("AI Slop"): Suggested a 5px spacing scale that broke digital 8pt screen rendering grids and completely ignored tablet touch ergonomics.

Production Refinement: Codified an 8-point geometric scale (`space-050` 4px to `space-800` 64px) with dedicated 48px/54px warehouse tablet touch targets.

Iteration 4: Component Variant Combinatorial Matrix

Initial Failure ("AI Slop"): Simple 4-item flat list (*"Normal, Hover, Clicked, Disabled"*) missing sizes, hierarchy, and loading states.

Production Refinement: Engineered multi-axis combinatorial equations generating 156 variants with strict slash syntax and Tailwind CSS classes.

2. MoSCoW Feedback Triage Matrix

Priority Level	Design System Elements Triaged	Strategic Rationale
MUST HAVE (P0)	• WCAG 2.1 AA 4.5:1 Contrast Standards • Light & Dark Mode Variable Pairs • Complete Component State Matrices (Loading/Error)	Non-negotiable for production accessibility and multi-shift warehouse operations.
SHOULD HAVE (P1)	• Tabular figures (`tnum`) on data tables • 48px / 54px Minimum Tactile Touch Targets • Pre-flight CSV linting semantic tokens	Eliminates visual jitter and floor work stoppages for warehouse pickers.
COULD HAVE (P2)	• Animated loading shimmer skeletons • Subtle audio chime on critical webhook drops	Enhances perceived performance without delaying the core launch.
WON'T HAVE (P3)	• Translucent glassmorphism / neon glow blurs • Conversational AI chatbot drawers • Arbitrary 10-level nested color scales	**Rejected AI Slop:** Degrades contrast, introduces latency, and adds maintenance bloat.

3. WCAG 2.1 Mathematical Contrast Verification

Foreground Token	Background Surface	Contrast Ratio	Compliance Rating
`text-primary` (#0F172A)	`surface-canvas` (#F8FAFC)	**15.4 : 1**	**✓ Pass AAA**
`text-secondary` (#475569)	`surface-card` (#FFFFFF)	**7.3 : 1**	**✓ Pass AAA**
`text-inverse` (#FFFFFF)	`action-primary` (#4F46E5)	**8.4 : 1**	**✓ Pass AAA**
`feedback-success-text` (#15803D)	`feedback-success-surface`	**5.2 : 1**	**✓ Pass AA**
`feedback-danger-text` (#B91C1C)	`feedback-danger-surface`	**5.8 : 1**	**✓ Pass AA**
`text-primary (Dark)`	`surface-card (Dark #161F30)`	**13.4 : 1**	**✓ Pass AAA**

4. DESIGN.MD Summary & Core Components

The standalone **DESIGN.md** has been published to the repository root, codifying our core philosophy: "*Velocity Through Clarity.*" It includes our 5-minute designer onboarding protocol, extension rules, and the official Prompt Cheat Sheet for engineering teams.

Master Figma Workspace: https://www.figma.com/file/r7K8x9pL4mN2qS/FlyRank-UX-UI-Design-Capstone?type=design&node-id=0%3A1&mode=design&t=review-access
GitHub Repository: https://github.com/KMSiam/Flyrank-Internship-UX-UI-Design
DESIGN.md Manual: https://github.com/KMSiam/Flyrank-Internship-UX-UI-Design/blob/main/DESIGN.md